

3. Determinants

3.1. Introduction to Determinants

Definition:

For $n \geq 2$, the determinants of an $n \times n$ matrix $A = [a_{ij}]$ is the sum of n terms of the form $\pm a_{1j} \det A_{1j}$, with plus and minus signs alternating where the entries $a_{11}, a_{12}, \dots, a_{1n}$ are fixed from the first row of A . In symbols:

$$\det A = a_{11} \det A_{11} - a_{12} \det A_{12} + \dots + (-1)^{1+n} a_{1n} \det A_{1n}$$

Given $A = [a_{ij}]$, the (i, j) -cofactor of A is the number given by:

$$C_{ij} = (-1)^{i+j} \det A_{ij}$$

Then

$$\det A = a_{11} C_{11} + a_{12} C_{12} + \dots + a_{1n} C_{1n}$$

Theorem 1:

The determinant of an $n \times n$ matrix A can be computed by a cofactor expansion across any row or down any column.

The expansion across the i -th row using the cofactor is

$$\det A = a_{i1} C_{i1} + a_{i2} C_{i2} + \dots + a_{in} C_{in}$$

The cofactor expansion down the j -th column is

$$\det A = a_{1j} C_{1j} + a_{2j} C_{2j} + \dots + a_{nj} C_{nj}$$

Theorem 2:

If A is a triangular matrix, then $\det A$ is the product of the entries on the main diagonal of A .

3.2. Properties of Determinants

Theorem 3: Row Operation

Let A be a square matrix

- If a multiple of one row of A is added to another row to produce a matrix B , then $\det A = \det B$
- If two rows of A are interchanged to produce B , then $\det B = -\det A$
- If one row of A is multiplied by k to produce B , then $\det B = k \det A$

The connection between A and U

Suppose a square matrix A has been reduced to an echelon form U . If there are r interchanges, then :

$$\det A = (-1)^r \det U$$

so:

$$\det A = \begin{cases} (-1)^r \det U = (-1)^r (\text{product of pivots in } U), & \text{when } A \text{ is invertible} \\ 0, & \text{when } A \text{ is not invertible} \end{cases}$$

Elementary Matrix Determinant:

If E is an $n \times n$ elementary matrix, then

$$\det E = \begin{cases} 1, & \text{if } E \text{ is a row replacement} \\ -1, & \text{if } E \text{ is an interchange} \\ r, & \text{if } E \text{ is a scale by } r \end{cases}$$

Theorem 4:

A square matrix A is invertible if and only if $\det A \neq 0$

Theorem 5:

If A is an $n \times n$ matrix, then $\det A^T = \det A$

Theorem 6: Multiplicative Property

If A and B are $n \times n$ matrices, then $\det(AB) = (\det A)(\det B)$

Linearity property of the determinant function

Suppose that the j -th column of A is allowed to vary, and write $A = [a_1 \ \dots \ a_{j-1} \ x \ a_{j+1} \ \dots \ a_n]$

Define a transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}$ by $T(\mathbf{x}) = \det[a_1 \ \dots \ a_{j-1} \ \mathbf{x} \ a_{j+1} \ \dots \ a_n]$

Then:

- $T(c\mathbf{x}) = cT(\mathbf{x})$, for all scalars c and all $\mathbf{x}$ in $\mathbb{R}^n$
- $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$, for all $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^n$

3.3. Cramer's rule, volume, and linear transformations

Theorem 7: Cramer's Rule

Let A be an invertible $n \times n$ matrix. For any $\mathbf{b}$ in $\mathbb{R}^n$, the unique solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$ has entries given by

$$x_j = \frac{\det A_j(\mathbf{b})}{\det A}$$

where $A_j(\mathbf{b})$ is the matrix obtained by replacing the j -th column of A with $\mathbf{b}$.

Theorem 8: An Inverse Formula

Let A be an invertible $n \times n$ matrix. Then:

$$A^{-1} = \frac{1}{\det A} \text{adj } A$$

so

$$A^{-1} = \frac{1}{\det A} \begin{bmatrix} C_{11} & C_{21} & \dots & C_{n1} \\ C_{12} & C_{22} & \dots & C_{n2} \\ \vdots & \vdots & & \vdots \\ C_{1n} & C_{2n} & \dots & C_{nn} \end{bmatrix}$$

Explanation of Theorem 8:

For an invertible $n \times n$ matrix, the j -th column of A^{-1} is a vector $\mathbf{x}$ that satisfies

$$A\mathbf{x} = \mathbf{e}_j$$

where $\mathbf{e}_j$ is the j -th column of the identity matrix. By Cramer's Rule, the i -th entry of $\mathbf{x}$ is $\frac{\det A_i(\mathbf{e}_j)}{\det A}$, which is $\frac{C_{ji}}{\det A}$ (since $\det A_i(\mathbf{e}_j) = C_{ji}$). So $A^{-1} = \frac{1}{\det A} [C_{ji}] = \frac{1}{\det A} \text{adj } A$

Theorem 9:

- If A is a 2×2 matrix, the area of the parallelogram determined by the columns of A is $|\det A|$
- If A is a 3×3 matrix, the volume of the parallelepiped determined by the columns of A is $|\det A|$
- Let $(\mathbf{a}_1)$ and $(\mathbf{a}_2)$ be nonzero vectors. Then for any scalar c , the area of the parallelogram determined by $(\mathbf{a}_1)$ and $(\mathbf{a}_2)$ equals the area of the parallelogram determined by $(\mathbf{a}_1)$ and $(\mathbf{a}_2 + c\mathbf{a}_1)$

Theorem 10:

Let $(T : \mathbb{R}^2 \rightarrow \mathbb{R}^2)$ be a linear transformation with standard matrix A . If S is a parallelogram in $(\mathbb{R}^2)$, then

$$\text{Area of } T(S) = |\det A| \times \text{Area of } S$$

Let $(T : \mathbb{R}^3 \rightarrow \mathbb{R}^3)$ be a linear transformation with standard matrix A . If S is a parallelepiped in $(\mathbb{R}^3)$, then

$$\text{Volume of } T(S) = |\det A| \times \text{Volume of } S$$